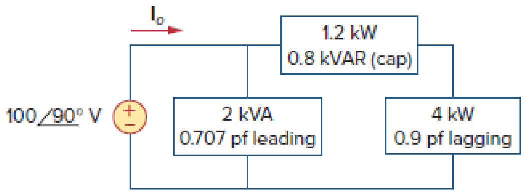
Chapter 11, Problem 61P

Problem

Given the circuit in Fig. 11.80, find *Io* and the overall complex power supplied.

Figure 11.80:



Step-by-step solution

Step 1 of 7

Refer to Figure 11.80 in the text book.

Draw the following block diagram.

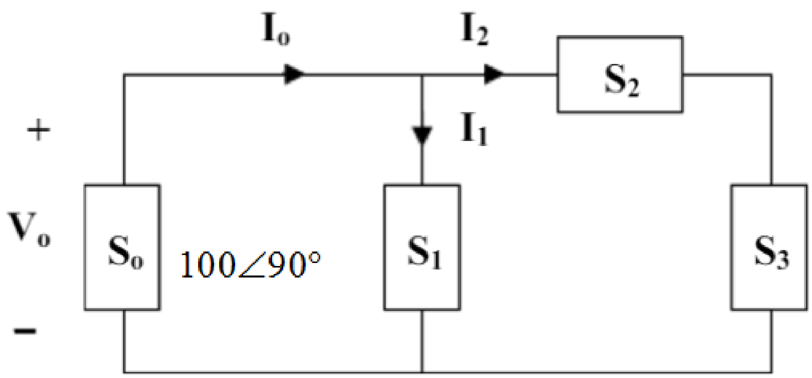


Figure 1

Step 2 of 7

Write the expression of complex power $\mathbf{S}_1$.

$$\begin{aligned}\mathbf{S}_1 &= P - jQ \\ &= P - j\left(\frac{P}{\cos\theta}\right)\sin\left(\cos^{-1}(\theta)\right) \\ &= 2\times10^3\times0.707 - j\left(\frac{1414}{0.707}\right)\sin\left(\cos^{-1}(0.707)\right) \\ &= 1414 - j1414.43 \text{ VA}\end{aligned}$$

Since power factor is leading, the reactive power is negative.

Write the expression of complex power $\mathbf{S}_2$.

$$\begin{aligned}\mathbf{S}_2 &= P - jQ \\ &= 1.2 - j0.8 \text{ kVA}\end{aligned}$$

Since power factor is leading, the reactive power is negative.

Step 3 of 7

Write the expression of complex power $\mathbf{S}_3$.

$$\begin{aligned}\mathbf{S}_3 &= P + jQ \\ &= P + j\left(\frac{P}{\cos\theta}\right)\sin\left(\cos^{-1}(\theta)\right) \\ &= 4 + j\left(\frac{4\,\text{kW}}{0.9}\right)\sin\left(\cos^{-1}(0.9)\right) \\ &= 4 + j1.937\,\text{kVA}\end{aligned}$$

Since power factor is lagging, the reactive power is positive.

The loads $\mathbf{S}_2$ and $\mathbf{S}_3$ are in series. Consider $\mathbf{S}_4$ as equivalent load.

$$\begin{aligned}\mathbf{S}_4 &= \mathbf{S}_2 + \mathbf{S}_3 \\ &= 1.2 - j0.8\,\text{kVA} + 4 + j1.937\,\text{kVA} \\ &= 5.2 + j1.137\,\text{kVA}\end{aligned}$$

Step 4 of 7

Determine the value of current $\mathbf{I}_2$.

$$\mathbf{S}_4 = \mathbf{V}_o \mathbf{I}_2^*$$

$$\begin{aligned}\mathbf{I}_2^* &= \frac{\mathbf{S}_4}{\mathbf{V}_o} \\ &= \frac{5.2 + j1.137\,\text{kVA}}{100\angle 90^\circ} \\ &= \frac{5323\angle 12.34^\circ}{100\angle 90^\circ} \\ &= 53.23\angle -77.66^\circ\ \text{A}\end{aligned}$$

Hence,

$$\mathbf{I}_2 = 53.23\angle 77.66^\circ\ \text{A}$$

Step 5 of 7

Determine the value of current $\mathbf{I}_1$.

$$\mathbf{S}_1 = \mathbf{V}_o \mathbf{I}_1^*$$

$$\begin{aligned}\mathbf{I}_1^* &= \frac{\mathbf{S}_1}{\mathbf{V}_o} \\ &= \frac{1414 - j1414.43}{100\angle 90^\circ} \\ &= \frac{2000\angle -45}{100\angle 90^\circ} \\ &= 20\angle -135^\circ\ \text{A}\end{aligned}$$

Hence,

$$\mathbf{I}_1 = 20\angle 135^\circ\ \text{A}$$

Step 6 of 7

Determine the value of $\mathbf{I}_O$ using Kirchhoff's current law.

$$\begin{aligned}\mathbf{I}_O &= \mathbf{I}_1 + \mathbf{I}_2 \\ &= 20\angle 135° + 53.23\angle 77.66° \\ &= 66.2\angle 92.4°\text{ A}\end{aligned}$$

Therefore, the value of current $\mathbf{I}_O$ is $\boxed{66.2\angle 92.4°\text{ A}}$.

Step 7 of 7

Determine the overall complex power supplied.

$$\begin{aligned}\mathbf{S}_o &= \mathbf{V}_o\mathbf{I}_o^* \\ &= (100\angle 90°)(66.2\angle 92.4)^* \\ &= 6.62\angle -2.4°\text{kVA}\end{aligned}$$

Therefore, the overall complex power supplied is $\boxed{6.62\angle -2.4°\text{ kVA}}$.